Personalized Learning Buddy

CLIENT REQUIREMENTS SPECIFICATION | ProofPay demo project

Document ID: PLB-SPEC-001 **Version:** 1.0 **Status:** Sample requirements for testing

This fictional document is designed to test ProofPay document loading, chunking, semantic retrieval, citations, and evidence-based acceptance criteria. The developer is expected to deliver a working responsive learning platform, not just static screenshots.

1. Project Overview

Build a web-based Personalized Learning Buddy for students preparing for mathematics and science examinations. The application shall adapt suggested learning material to individual progress and provide measurable practice feedback.

2. Target Users

Primary users are students aged 15-20. A teacher can review aggregate learning progress, but teachers must not see students' private journal notes.

3. Agreed Technical Scope

TECH-01 - Use React for the frontend and a Node.js API backend. PostgreSQL is the preferred persistent database.

TECH-02 - Support the latest stable versions of Chrome and Firefox on desktop and mobile-width browser viewports.

TECH-03 - Use a demonstration data set containing at least 30 learning resources across algebra, calculus, physics, and chemistry.

4. Deliverables

DEL-01 - A GitHub repository with application source code, a readable README, and documented setup instructions.

DEL-02 - A running demo build, a test report, and sample student accounts created only for testing.

DEL-03 - A brief architecture note describing the recommendation logic and how student data is stored.

Functional Requirements: Accounts and Learning Plans

5. Authentication and Student Profile

AUTH-01 - Students shall be able to register with email and password. Duplicate emails must be rejected with a useful error message.

AUTH-02 - Students shall be able to log in and log out. Invalid credentials must not grant access to a student dashboard.

AUTH-03 - A student profile shall store preferred subjects, estimated study minutes per day, and self-reported proficiency level: beginner, intermediate, or advanced.

6. Personalized Learning Recommendations

REC-01 - The Learning Buddy shall recommend exactly five learning resources on the dashboard after onboarding, ranked using the student's selected subjects, proficiency level, and the most recent quiz results.

REC-02 - When a student scores below 60% on an algebra quiz, the next recommended set must prioritize at least two beginner or remedial algebra resources.

REC-03 - After a student finishes a resource, it shall be marked completed and should not appear as an uncompleted recommendation.

REC-04 - The student must be able to see a short explanation of why each resource was recommended, such as "Recommended because you struggled with quadratic equations."

7. Daily Study Plan

PLAN-01 - Generate a daily study plan using the student's selected subjects and available study time. Default study time is 45 minutes per day if the student skips this choice.

PLAN-02 - Students can manually reorder their daily tasks, and the new order must persist after page refresh.

PLAN-03 - Display progress as completed tasks divided by total assigned tasks for the day.

Acceptance note: Recommendations must react to profile changes and quiz performance; a permanently hardcoded set of five resources does not satisfy REC-01 or REC-02.

Functional Requirements: Quizzes and Progress

8. Adaptive Quiz System

QUIZ-01 - Each quiz shall contain ten multiple-choice questions drawn from the chosen subject and difficulty level.

QUIZ-02 - After quiz submission, show the score as a percentage, identify incorrect answers, and provide short explanations for corrections.

QUIZ-03 - If the score is 80% or higher, the next recommended quiz difficulty may increase by one level. If the score is below 60%, the next quiz should remain at the same or a lower difficulty.

QUIZ-04 - Quiz answers must be saved so that a student can view the last five completed quiz attempts.

9. Progress Dashboard

PROG-01 - Show a weekly bar chart of study minutes for the previous seven calendar days.

PROG-02 - Display subject-level mastery as a percentage computed from the latest three quizzes in each subject; if fewer than three quizzes exist, use the available results and label the count.

PROG-03 - Show the current study streak as the number of consecutive calendar days with at least one completed study task.

10. Notifications

NOTIF-01 - Show an in-app reminder when the student has no completed tasks for the current day and the chosen study window begins.

NOTIF-02 - Email and SMS reminders are outside the agreed scope of this milestone.

11. Accessibility

ACCESS-01 - All core flows must be operable with the keyboard, with visible focus indicators and descriptive form labels.

ACCESS-02 - The application must present readable content at 375 px viewport width without horizontal page scrolling.

Milestones and Acceptance Tests

12. Payment Milestones

Milestone	Deliverables	Agreed amount
M1 - Foundation	Authentication, profile, dashboard shell	$150
M2 - Learning Engine	Recommendations, daily plans, resource completion	$250
M3 - Assessment	Quizzes, progress charts, accessibility review	$200

The client reviews AI-generated verification evidence and may approve, request revisions, or reject any milestone. Automated AI scoring alone must not authorize payment.

13. Concrete Acceptance Test Scenarios

TEST-01 - Register two accounts using the same email. The second attempt must be rejected (AUTH-01).

TEST-02 - Log in with an incorrect password. The dashboard must remain inaccessible (AUTH-02).

TEST-03 - Set algebra proficiency to beginner and submit an algebra quiz with 40% score. At least two of the next five recommended resources must be remedial algebra material (REC-02).

TEST-04 - Set available study time to 30 minutes, reorder two tasks, refresh the page, and confirm the order remains unchanged (PLAN-02).

TEST-05 - Submit a ten-question quiz with eight correct answers and verify the displayed result is 80% and correction explanations are shown (QUIZ-01/02).

TEST-06 - Open the application at a 375 px browser width and verify the main dashboard does not require horizontal scrolling (ACCESS-02).

14. Submission Evidence

For each milestone, the freelancer must submit a GitHub repository URL, pinned commit SHA, README setup instructions, and at least one screenshot or short screen recording demonstrating the milestone. Existing unit tests, integration tests, and Playwright reports should be included when available.

Quality Criteria, Constraints, and Open Questions

15. Non-Functional Requirements

PERF-01 - On the provided demo data set, the main dashboard should reach usable interactive state within three seconds on a typical broadband connection and a recent laptop. Measurement method must be documented.

SEC-01 - Student passwords must never be stored in plaintext. Protected student endpoints must require authentication.

PRIV-01 - A user must not be able to read another student's quiz history by modifying a URL or API request parameter.

RELIAB-01 - API errors must show a useful message and must not cause the entire application to crash.

16. Explicitly Out of Scope

OUT-01 - No live video tutoring, audio calls, or AI-generated avatars are required.

OUT-02 - No paid subscription billing, payment gateway, or cryptocurrency support is required inside the Learning Buddy application.

OUT-03 - No native Android or iOS application is required; a responsive web app is sufficient.

17. Intentionally Unspecified Details

The document does **not** specify the application's logo color, whether dark mode is required, which LLM vendor to use, the geographic hosting region, or whether parents should receive weekly emails. The requirements agent should explicitly say these items are unspecified rather than inventing them.

18. Review and Final Approval

The client judges each criterion based on demonstrable evidence. Any untestable requirement must be marked UNVERIFIED rather than PASS. Milestone approval is a separate client action; a tip is optional and not predetermined in this document.

Document purpose: Synthetic sample for retrieval evaluation. No real client, private account, or payment is represented.